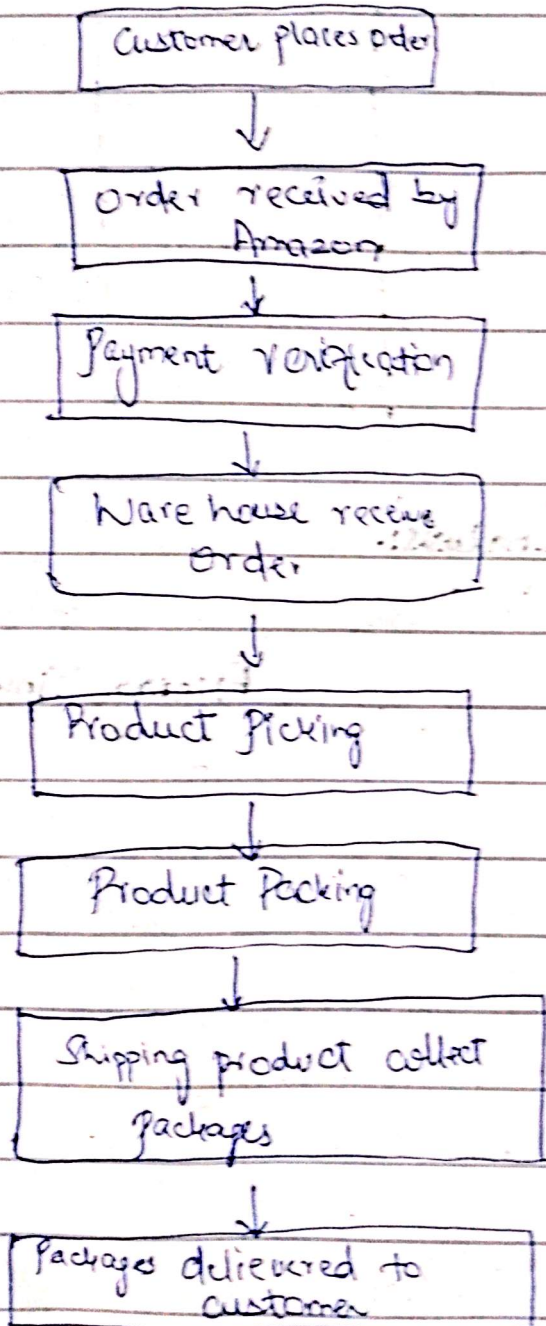


Sayim Khan

FA22-BSE-049

Company 1:- Amazon

Process Flow:-



Flow Analysis:-

	Activity	Value Added	Possible Delays
1	Customer place order	Yes	No
2	Payment verification	Yes	Payment failure
3	Warehouse receive request	Yes	System delay
4	Picking product	Yes	Worker travel time
5	Packing	Yes	Manual picking error
6	Shipping	Yes	Traffic / weather
7	Delivery	Yes	Wrong address

Quantitative Analysis:-

Suppose Amazon receives 10,000 orders/day

Activity	Average Time
Order processing	2 mins
Payment verification	1 min
Picking	8 mins
Packing	1 min
Shipping	4 mins
Delivery	24 hours

Total time = 20 minutes

Efficiency Calculation:-

Actual working time = 20 mins

Total Customer waiting time = 24 hrs
= 1440 mins

Process Efficiency

$$= \frac{20}{1460} \times 100 = 1.37\%$$

Bottleneck:-

- Warehouses workers walk long distance
- Manual packing slows operations
- Traffic causes delivery delays.
- Stock shortages delay packing.

Improvement Opportunities:-

- * Use warehouse robots.
- * Automatic packing machines.
- * AI demand forecasting.
- * More local fulfillment centers.
- * Barcode scanning at every stage.

Expected Improvements:-

Before	After
Picking = 8 mins	5 minutes
Packing = 5 mins	3 mins
Processing = 20 mins	15 mins

$$\text{Efficiency improved} = \frac{20-15}{20} \times 100$$

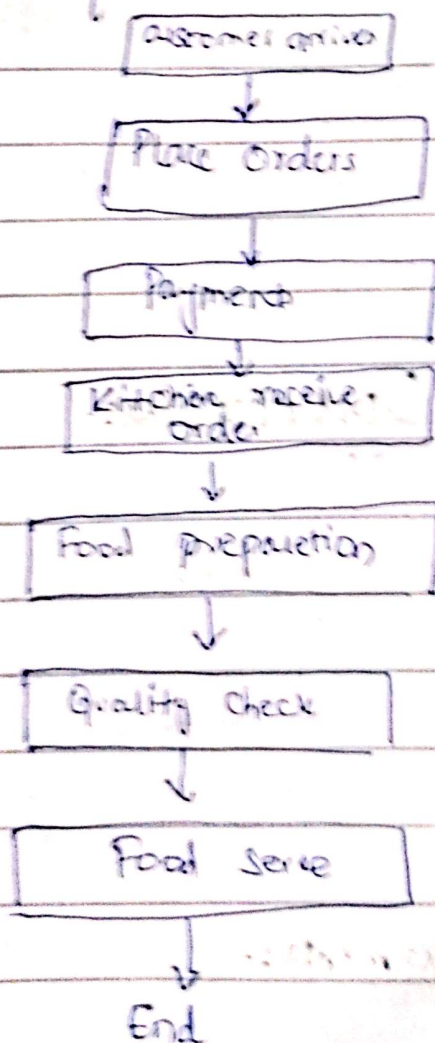
$$= 25\%$$

Company 2 : McDonald

Business Process:

Customer food ordering process

Process flow:-



Flow Analysis:-

1.	Customer order		Yes		Queue
2.	Payment		Yes		Cash handling
3.	Kitchen receive order		Yes		Communication
4.	Cooking		Yes		Food preparation
5.	Quality check		Yes		Inspection
6.	Serving		Yes		Crowded counter

Quantitative Analysis:-

Ordering		2 min
Payment		2 min
Kitchen receives order		1 min
Cooking		8 min
Quality check		1 min
Serving		2 min

Total time = 16 minutes.

Bottle neck:-

- Long queue during lunch.
- Cash payment increase payment time.
- Kitchen staff overload.

Improvements:-

- Self service kiosks.
- Digital payments.
- Better staff scheduling.
- Prepare frequently order items in advance.
- Digital kitchen display system.

Expected Improvements:-

Before	After
Order time = 2min	1
Payment = 2min	1
Cooking = 8min	6

= 8 mins

$$= \frac{16 - 8}{16} \times 100 = 50\%$$